

ABS Level I Basic

ABS Core Batch

Version 2.5

Agenda

- 1 Features of ABS Batch
- 2 Central DB server
- 3 ABS operating day
- 4 Processing during the batch
- 5 Architecture of the batch applications
- 6 Summary



Contents of the module

What can you expect in this unit?

You get an insight:

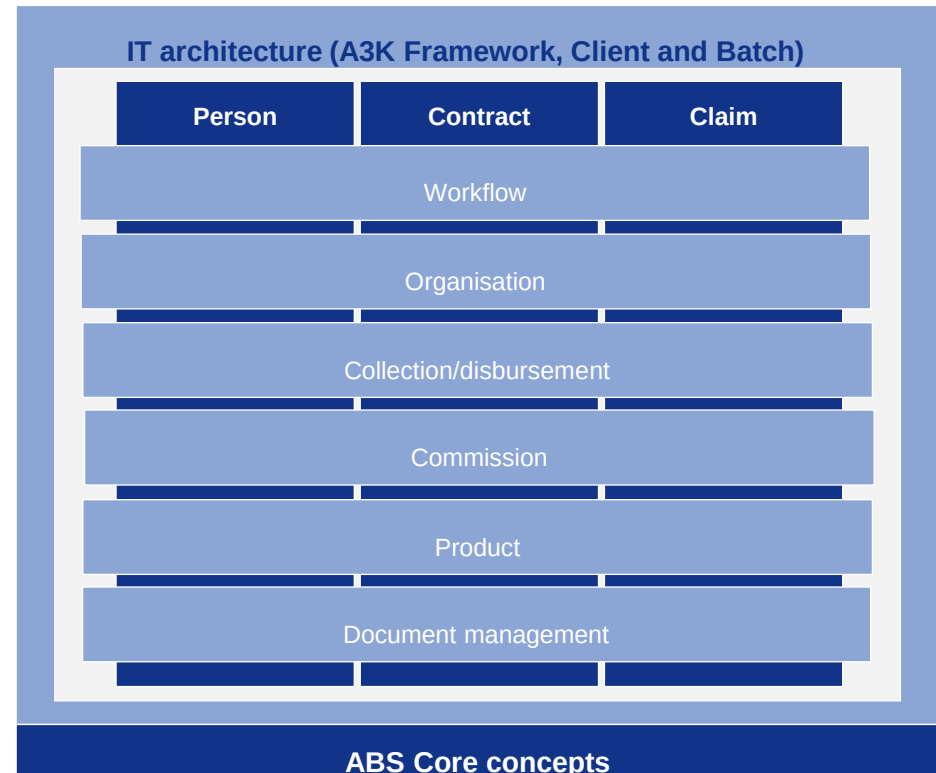
In the role of the “Host” in ABS

In central data management

In the ABS operating day

Into batch processes exemplary on basis of applications

In the architecture of the ABS batch services



Batch is responsible for automation and post-processing of processes in ABS
- No online without batch - The operationally leading inventory is managed on the Host.

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ABS batch process

ABS batch process takes place on an IBM mainframe and since 2013 also on a Linux server

Daily, large volumes of data are delivered to ABS Host by:

- Clerks in the ABS Client
- Customer and broker with web services
- Experts
- Claims sales representatives
- Business partners
- Documents from the mail receipt system
- Customer calls



All data are checked, saved, compressed, linked to each other and processed further

What is batch processing for?

Was batch processing not already replaced by modern IT concepts long ago?

To start EDP, punched card decks - **a batch** - were imported and processed card by card.



An increasing number of functions for managing contracts, claims or customer data can be completed directly in a client or a portal. **Downstream process steps** (accounting, commission, payments, evaluations) are initiated there and then executed in the background.

Even today, the IT of an insurance company needs a technical infrastructure and application systems to deal with the **large data volume** efficiently and reliably in a batch mode.

ABS therefore supports the **24x7 hours service** with **thousands of batch programs** and jobs on the IBM mainframe or high-performance servers under Linux in a highly available computer centre.

Distinction between batch job and batch services

Explanation of frequently used terms in the ABS batch environment

Program

A program or application summarises all necessary components in order to execute an automatic processing with previously defined processes.

Technical programs underlie functional business rules.

Batch job

A job starts a program in a single instance and processes record by record.

- No parallelism
- For programs with less input or e.g. need-based carrier.

Batch service

A batch service is a particular batch job and provides amongst other things the following benefits:

- Supports parallel activities
- Can be defined as a permanent carrier
- Can be controlled externally

Process

A process is a bundle of batch components for processing of functional context.

ABS batch processes are “state of the art”

ABS batch applications are developed for a reliable 24x7 operation

Scalable

Flexible adjustment to the workload:

- Higher through-put e.g. in year-end business
- Economical computer utilization on a “calm afternoon”

Error-tolerant

Individual applications or claims are escalated in the case of an error. Concerned persons are informed quickly.
The system continues to run as stable without manual interventions

Parallel processing of online and batch

By using the work tables, the batch programs relieve the central database for online access of ABS clients

Uniform architecture

Good maintainability of batch services through a uniform generated program framework.

ABS processes are not “shadow processing”

How can you see what is happening on the host?

Online monitoring

Work volume, work progress, escalations and errors are monitored continuously

Controls of the system resources

The SMF (System Management Facility) continuously collects measurement data for controlling and planning computer utilization

Logging in DB2

Grand totals, computer utilization and runtimes of ABS processes are stored for controlling and evaluating a database

Application and system side there are ways to monitor batch services and to control them based on workload.

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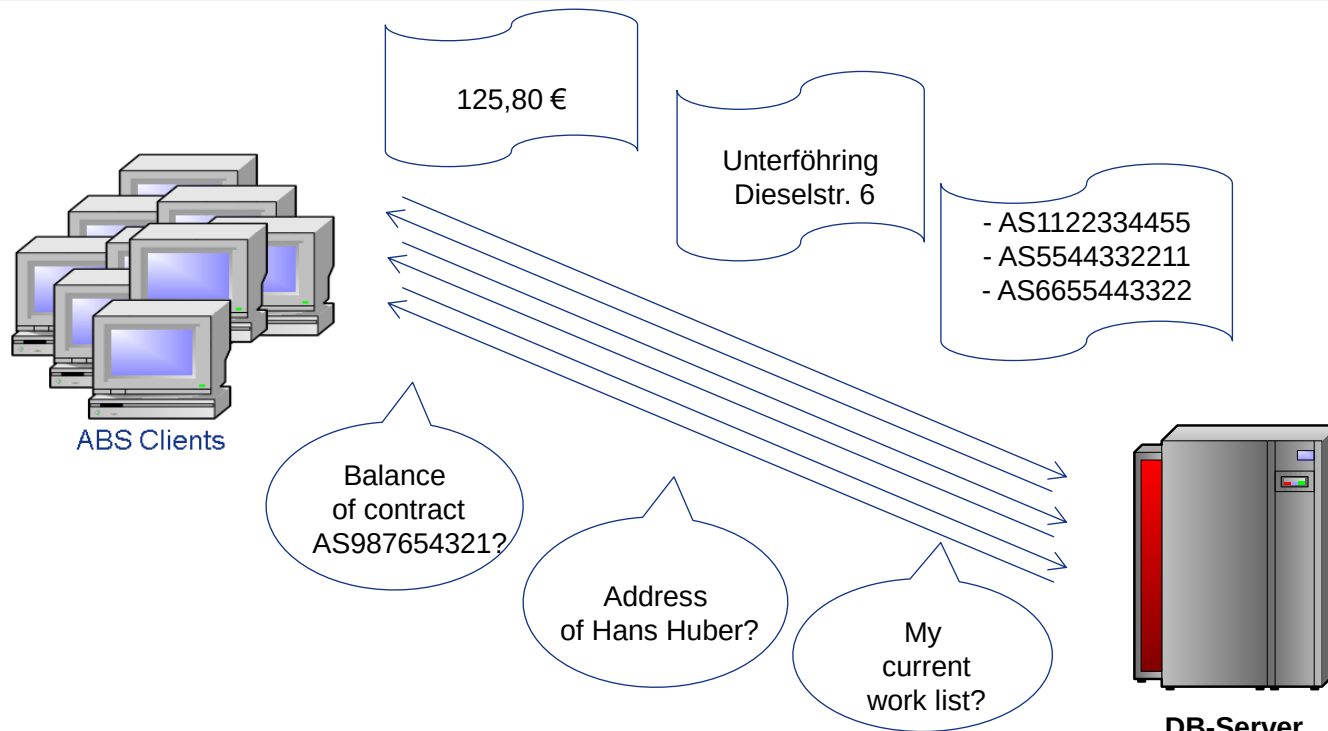
Direct read access on to the database server

Targeted individual queries for the desired information.

ABS Clients read data records directly on the Host database

This means:

- Many parallel accesses - high system load
- Long response times



Grouping of database access operations in a service component of the database management system

In addition to the standard SQL, stored procedures are also called to optimise the performance

Stored Procedures

Frequently used instructions, which require many individual commands, are summarised in a stored procedure and executed on the database server with a single call.

Complex database queries (e.g. routing rules)

Execution of bookings

Allocation of new policy numbers

Save documents of Client (e.g. PDF files) in the database as “Blob” (Binary Large Objects)

Workload management optimises response times and system load

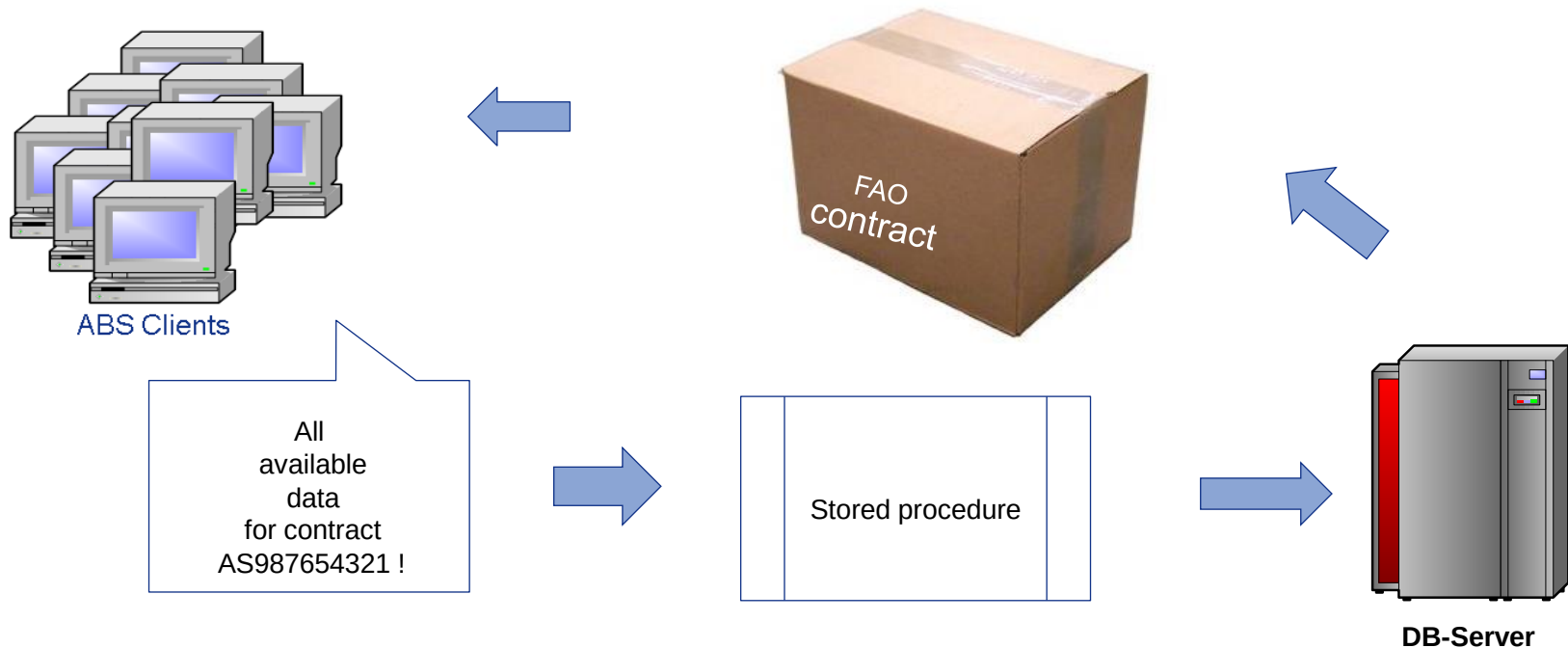
Read access with stored procedure on the database server

Provision of bundled information

The ABS Clients request complete data for one or several data objects
(bulk read)

This means:

- Lower system load
- Good Client performance



Write access with stored procedure on the database server

The Client may only change the database via the host

All changes and inclusions are transferred by the Client to a stored procedure on the host and executed there centrally

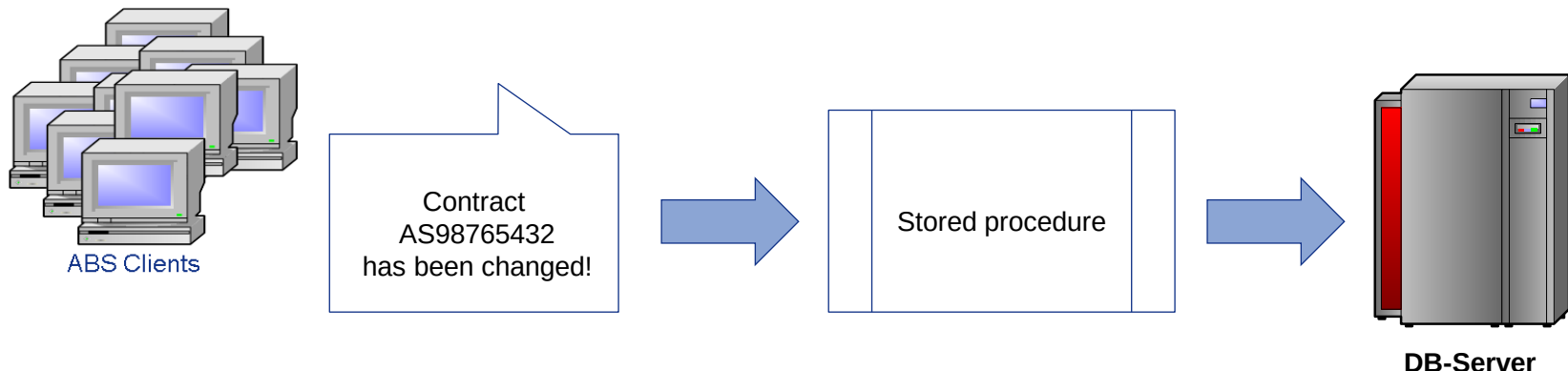
All SQL statements for a transaction are recorded and archived in TPROTOKOLL

In TVERAENDERUNG, new content is collected for forwarding

The business object manager (see next slide) is called

Work lists are updated, if after a change processing is to be initiated by the next clerk

Customer exit can be called, in which further processing steps are executed



Why must business object manager be used?

This core component guarantees the integrity of the ABS database

Basic rule

All DB changes to the operative tables must be documented comprehensibly and tamper-proof.

Business object activity

For every change to a business object from online or Batch type of activity, time and user or program are stored in the table TGSCHFAKT.

Modification log

Insert / update / delete for each table, which is affected in a business object activity, in TVERPROTOKOLL
optional: Link to historical version before the change

History management

For all tables, it can be defined in the configuration table TDBBERBESCHR that the historical status should be managed.

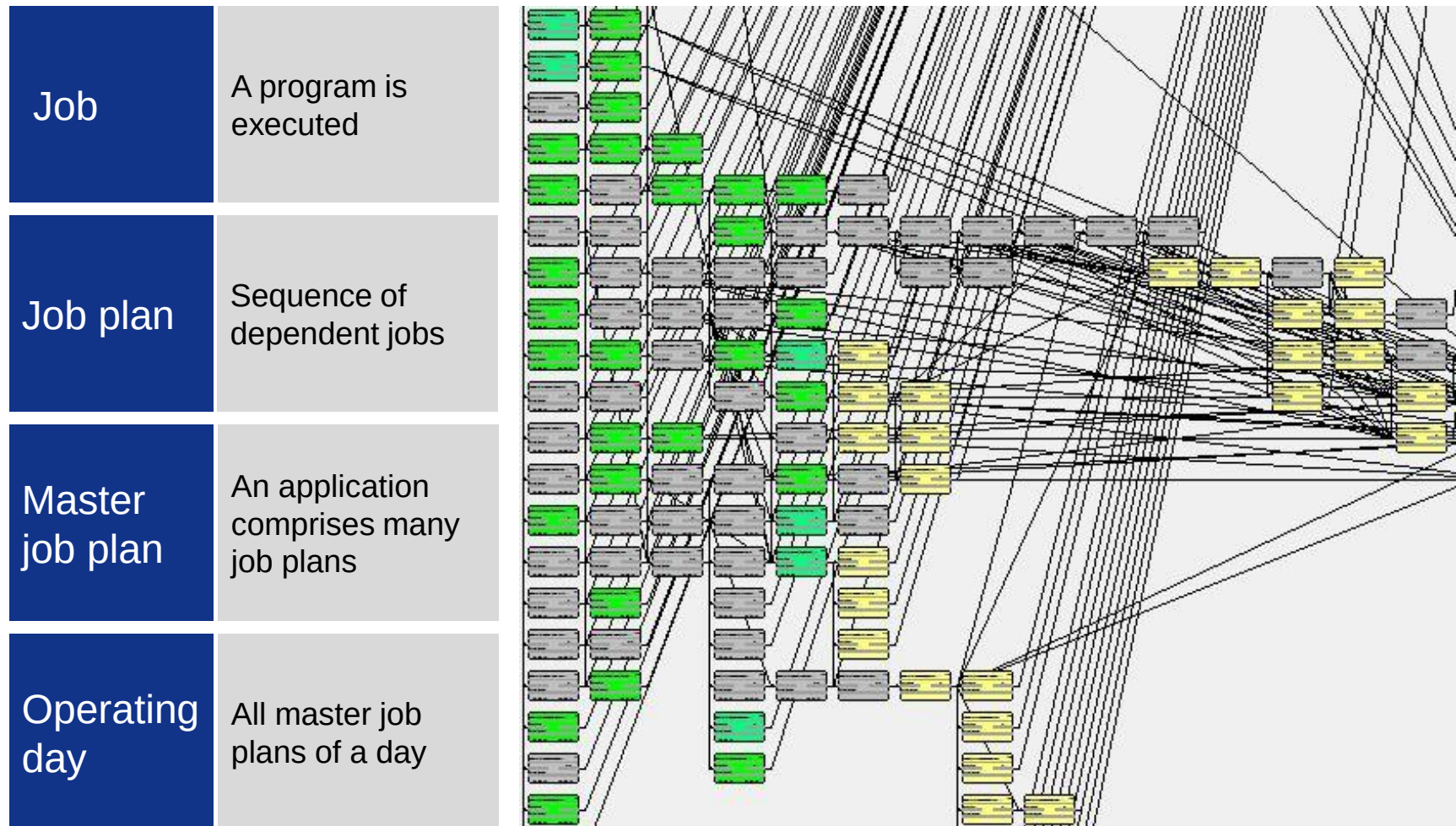
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On the ABS operating day, a complex network of jobs is executed on the host

ABS operating day determines the time sequence of the daily applications



ABS operating day

Specific basic conditions must be observed to coordinate the time sequence

The most important basic conditions

Start of operating day approximately 00:30 hours

Start and end of jobs are not rigidly bound to points in time but are linked to successful processing of upstream processes

Target duration corresponds to a real day

Different operation on weekends, holidays and in case of special actions



ABS operating day - features

ABS operating day is the mapping of a logical working day within the ABS application.

The most important features

Start

Infrastructure for ABS operation is provided, this means to save and cleanup of work tables, checking ABS READ/WRITE mode.

Special days

Cyclical events (e.g. holidays, weekends) are observed

Process

The sequence of jobs and their dependencies is provided by a planning software, started and monitored.

End

The database is saved.

The logical ABS operating day is not necessarily a calendar day.

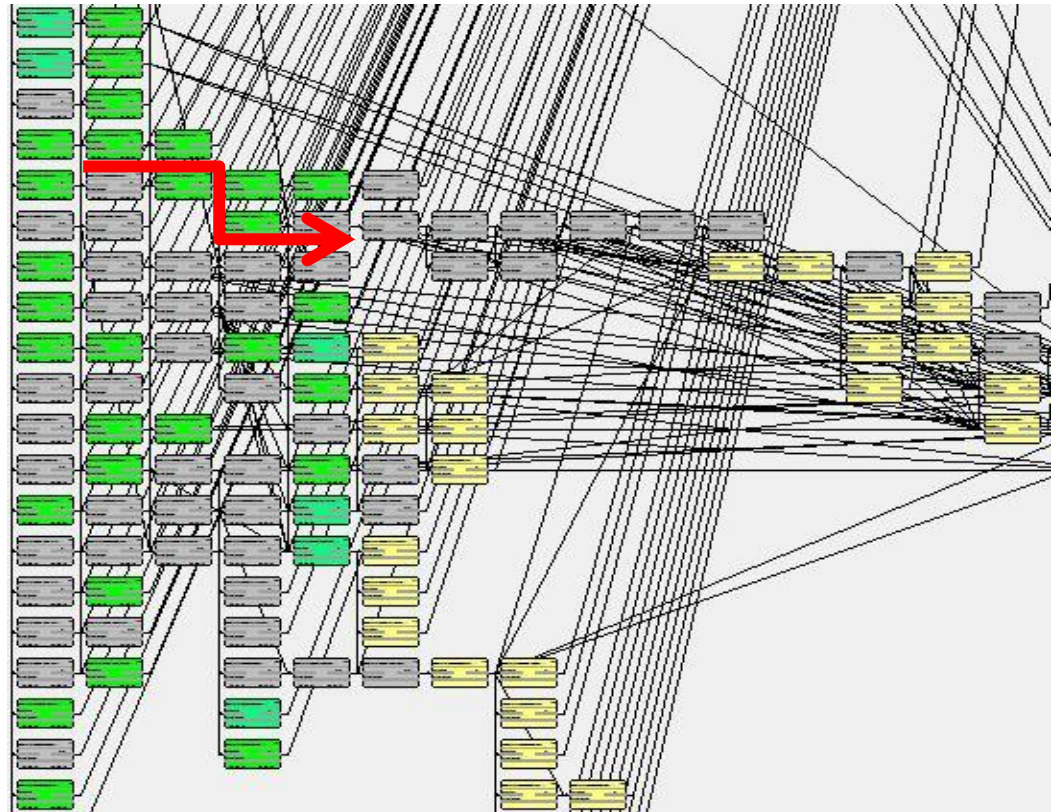
ABS operating day - “critical path”

The “critical path” is a sequence of jobs, which must be successfully completed on this logical working day

“Critical” jobs are identified and monitored accordingly

On cancelling a critical job,
no subsequent jobs are
executed within this
process or application

It is the operating day! The
responsible employees are
informed immediately



“Critical path” of the ABS operating day - use cases

Essential tasks for a successful ABS operating day

Examples of critical jobs:

Provision of environment variables (e.g. logical working day)

System / infrastructure consistency checks (DB2-system, web server, etc. are accessible?)

Mandatory processing of impulses of TBATCH_PROT or TDRUCKOUTDATEN (depends on logical working day)

Cash flow is carried out externally as well as internally

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Batch processes supporting online operations

ABS operating day distinguishes between online and offline operation

Preparation for the online operation

Reminder	Checking the balance and coverage of all contracts
Work list	Work lists are created for the clerk; process slip and scheduled tasks are created
Collection/disbursement	Incoming payment: Updating the contract balance

Batch processes parallel to online operation

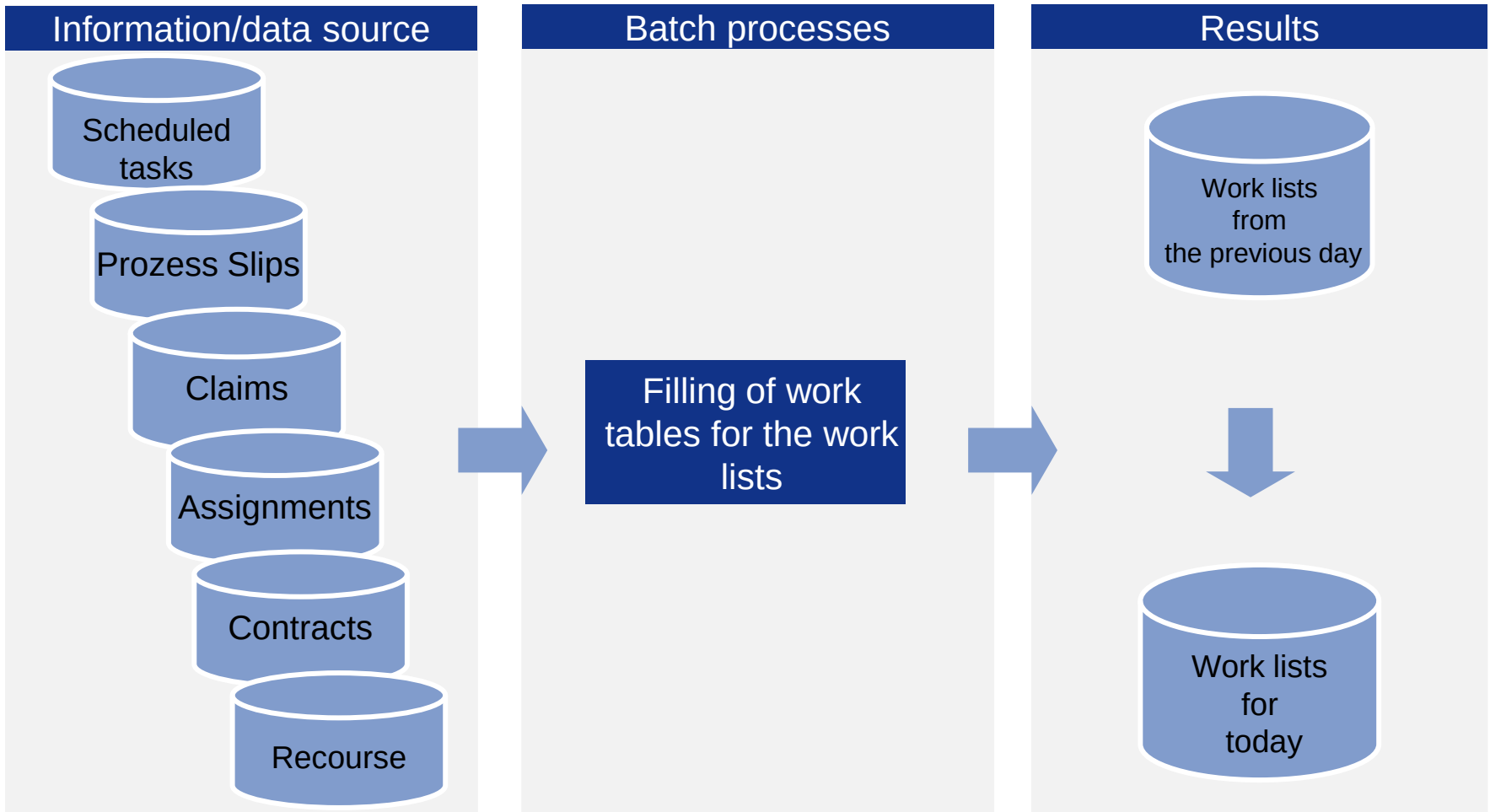
Policy processing	Accepting new and changed contract data
Claim post-processing	Checking and settling notices of loss
Settlement interface	Accounts department delivers collection/disbursement

Post-processing

Collection/disbursement	Direct debit, credits etc. from policy processing and claims
Output management	Printed documents from policy processing, accounting and claim
Evaluations	Provision of data warehouse etc.

As preparation for the online operation work lists are created.

The entire database is searched in order to determine the current workload



The work lists are filled in the host by analysing the operational tables.

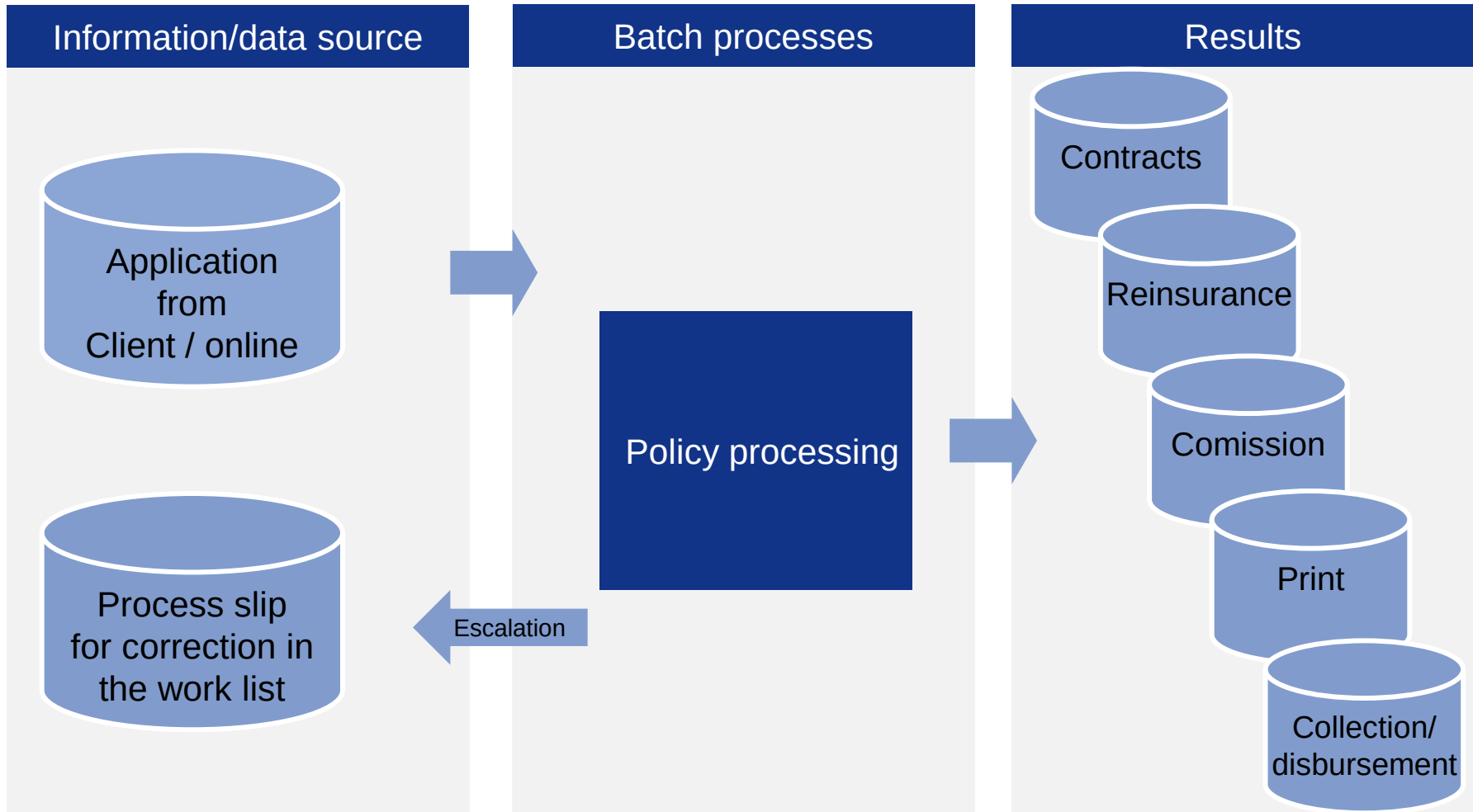
Examples of work lists filled by the host

Only a small selection is shown here. There are a number of wordlists for each working range

Claim work list	Claims assigned to a clerk or team
Application work list	New and to be changed applications
Assignment work list	Assignments in claim processing
Reminder evidence work list	Status of contracts in reminder
Document work list	Incoming documents, which are to be assigned to a contract or claim and are not functionally distinct

Processing parallel to online - policy processing

Contracts are created from changes and new applications



Continuous acceptance of work orders of ABS Clients and renewed provision of current results

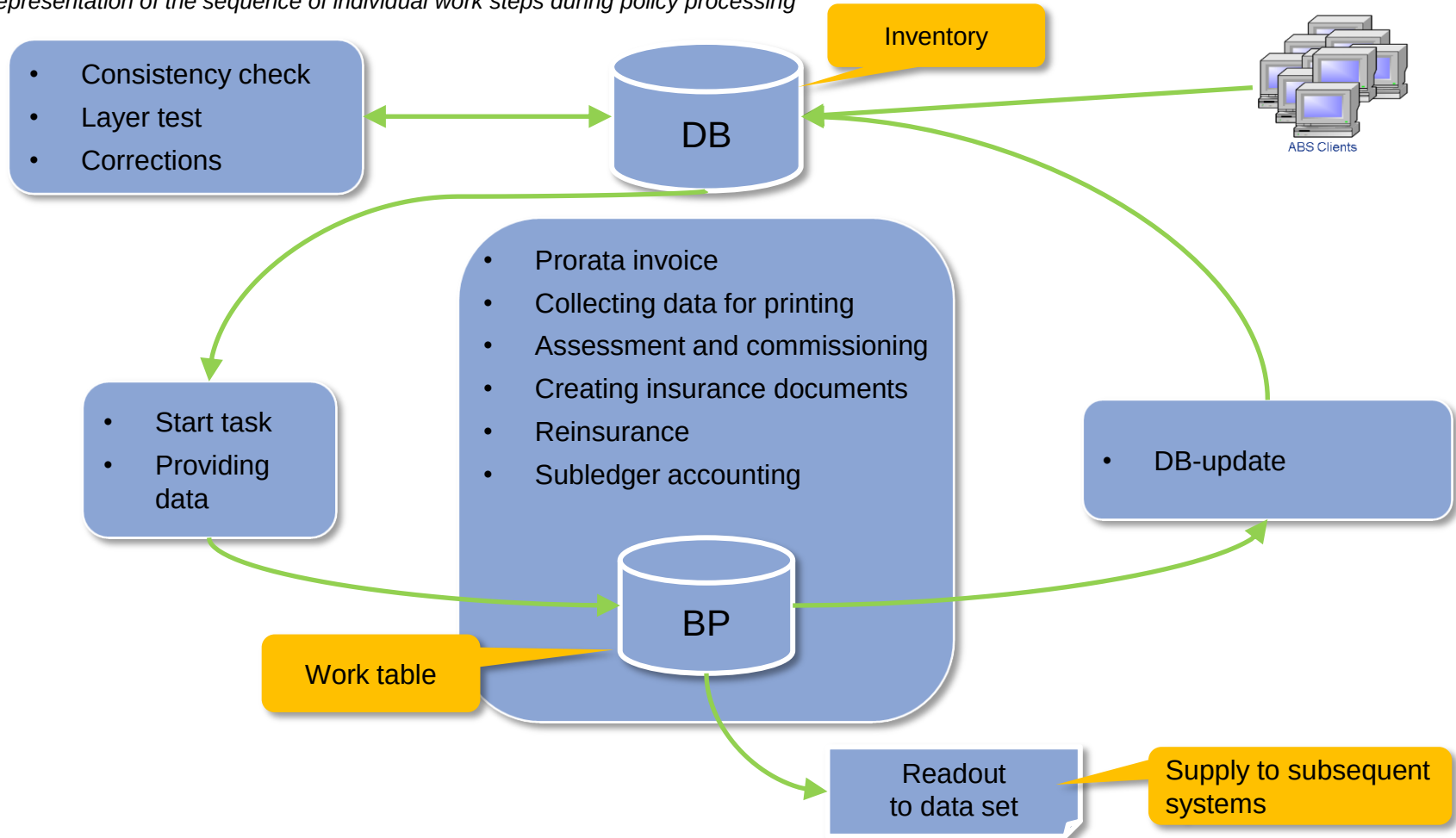
Processes of policy processing during batch processing.

Functional process on the host for final processing of an application

Continuous checking and correction of inputs	Continuous correction and if required creation of process slip
	Error information to the clerk
Processing an application or contract amendment	Prorata calculation
	Commissioning
	Collect print data, creating printed documents
	Booking of calculated amounts to the relevant accounts (contract, commission, customer account or NCP account)
Completion	Database update

Policy processing is divided into several work steps.

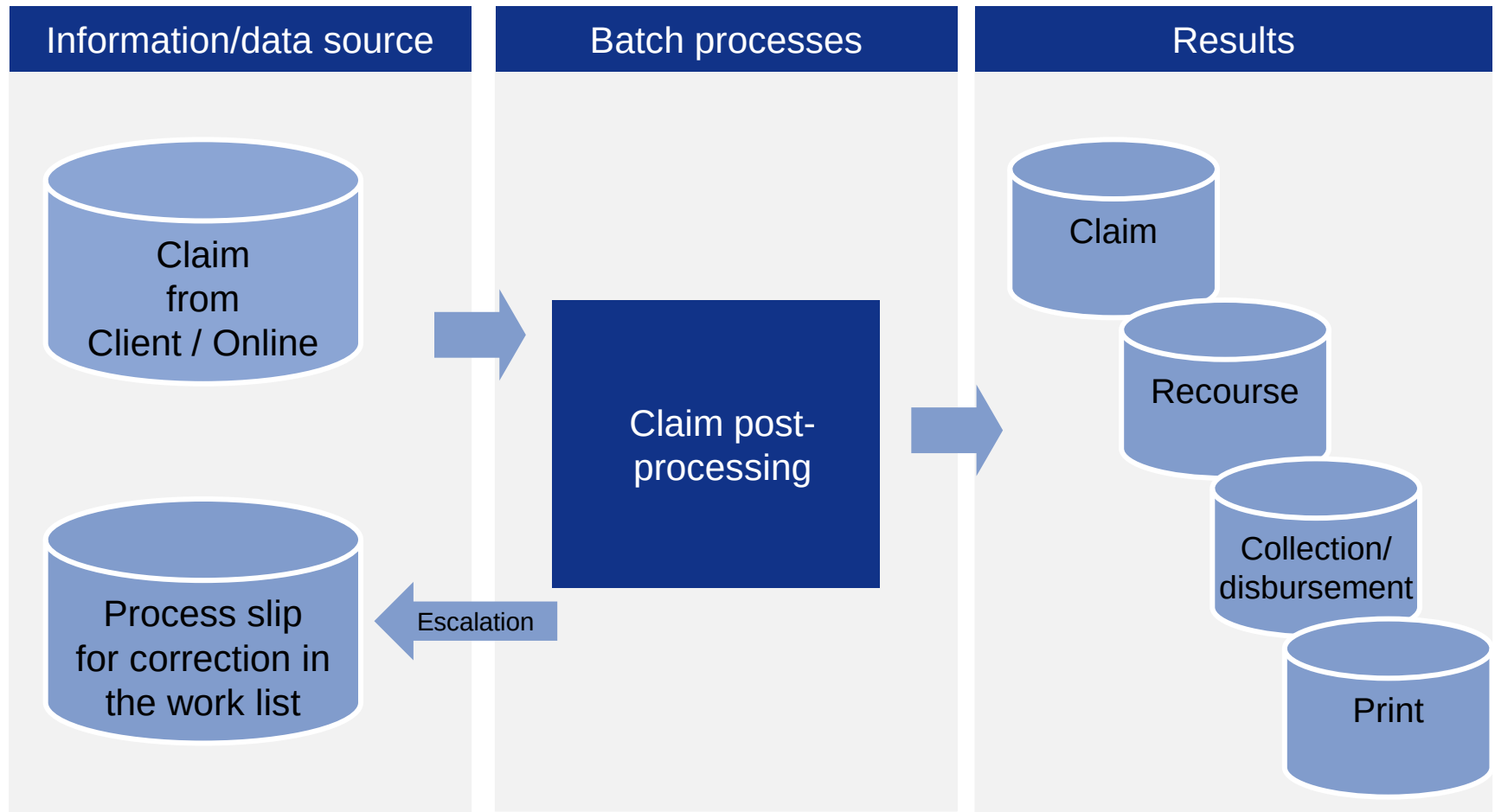
Representation of the sequence of individual work steps during policy processing



Basic work steps are specified by ABS Core and cannot be changed. However, customer-specific extensions are possible.

Processing parallel to online - claim

Finished claims from online are continuously fully processed in the host



Continuous acceptance and processing of reported claims as well as renewed provision of current results

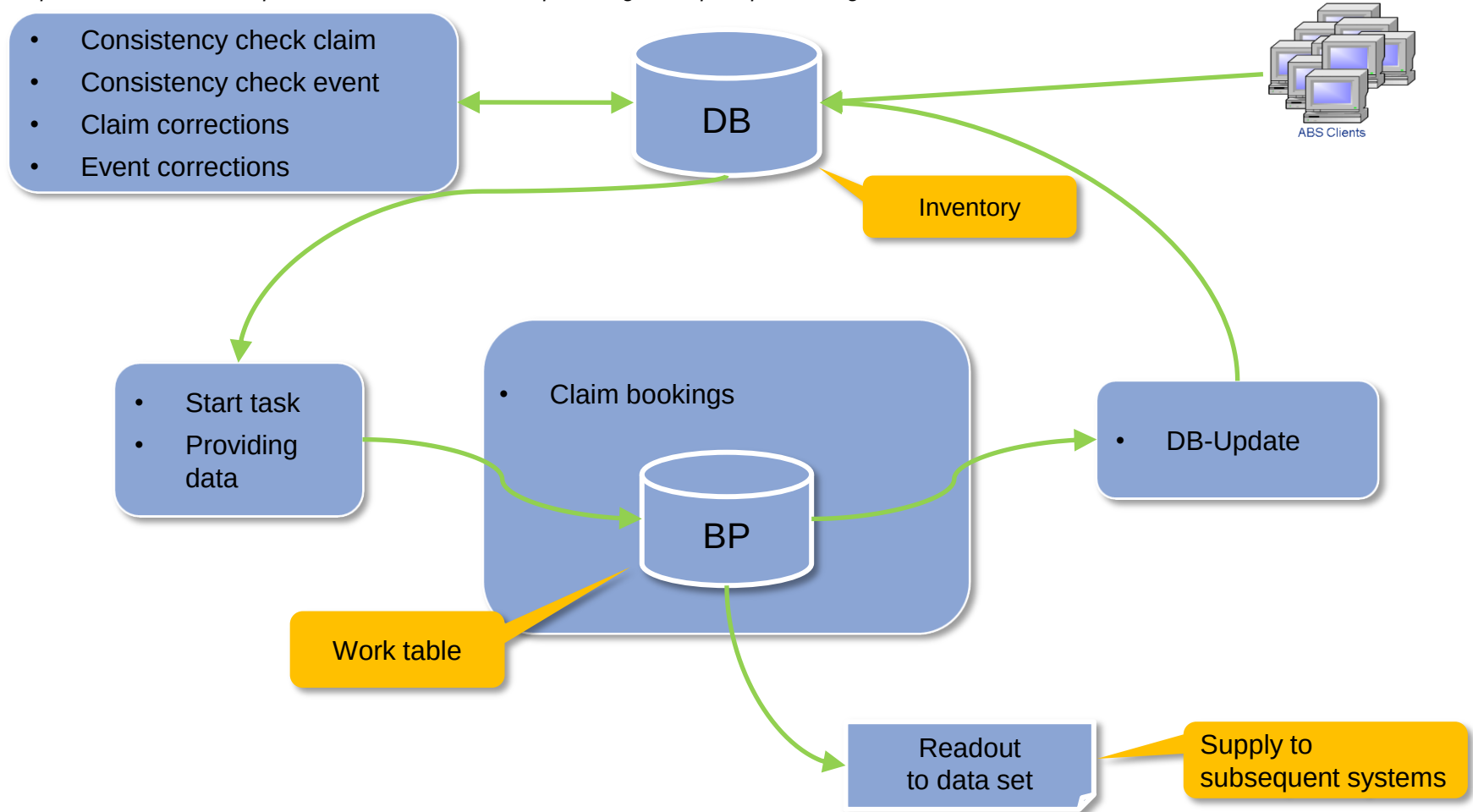
Batch processes in claim post-processing

Functional process on the host for final processing of a claim

Continuous acceptance of claims and events from online processing	Consistency check of claims (e.g. suspicion of double claim – escalate with a process slips in case of a problem)
Processing a claim	Checking excess, incoming and outgoing payment as well as recourse agreements
	Executing claim bookings; filling of interfaces for the general ledger; netting the bookings
	Filling the co-insurance interfaces (lead office and co-insurance business)
Completion	Executing claim resolutions (setting closing date, setting provision to 0)

Work steps of claim post-processing

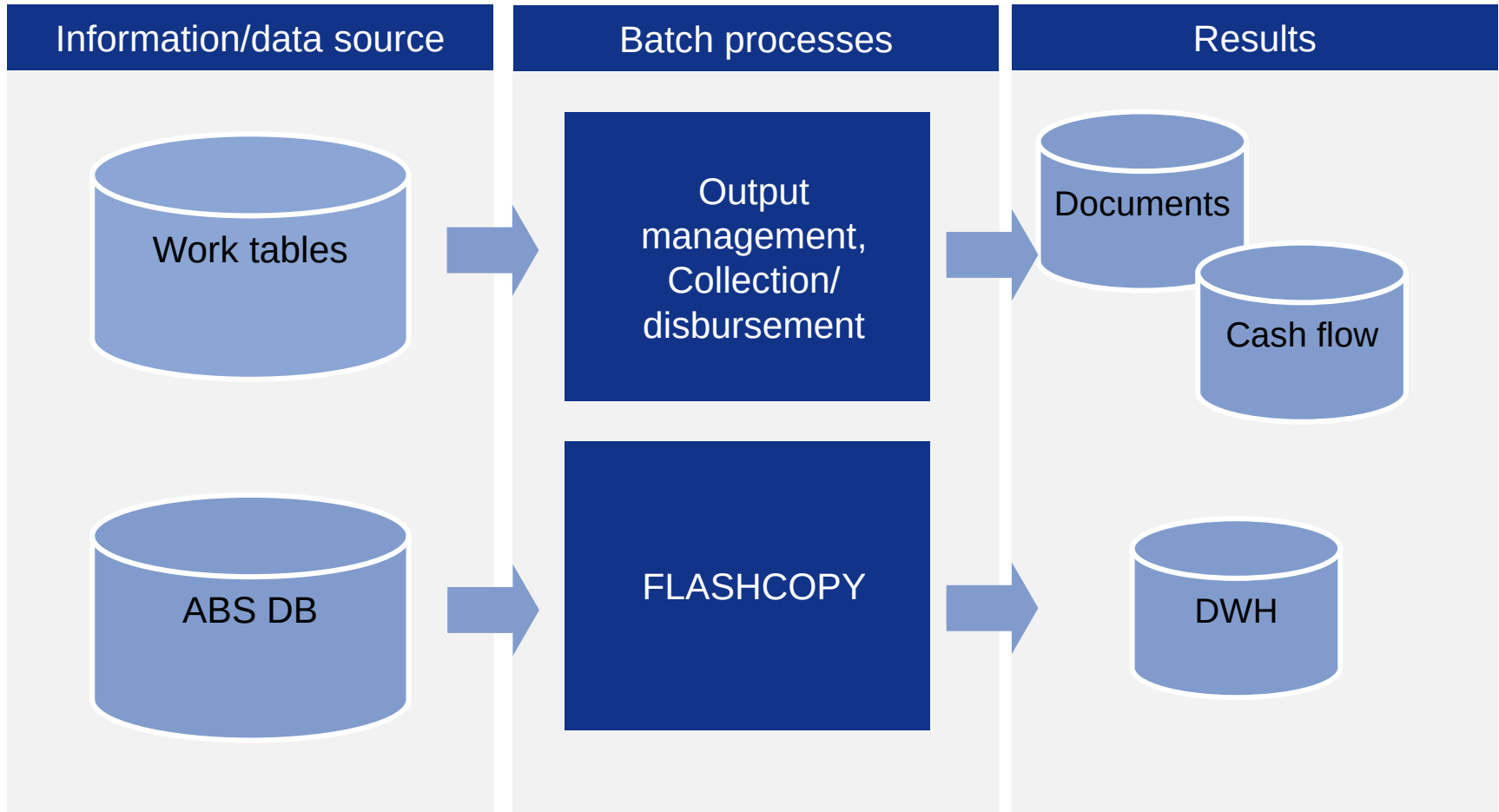
Representation of the sequence of individual work steps during claim post-processing



Basic work steps are specified by the ABS Core and cannot be changed. However, customer-specific extensions are possible.

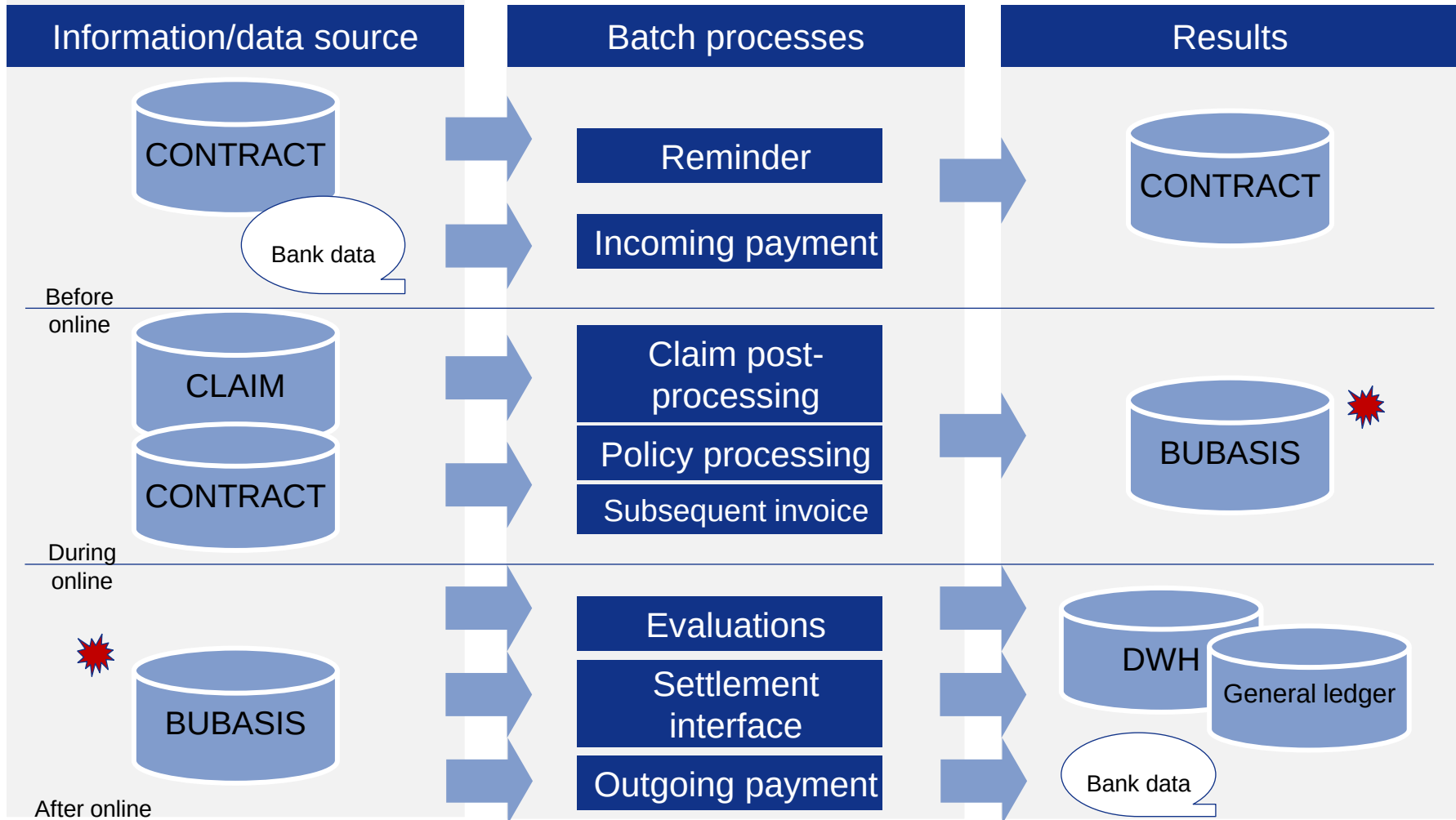
Batch processes after online

Online results are processed , compressed and provided to subsequent processes.



Batch functions in collection/disbursement

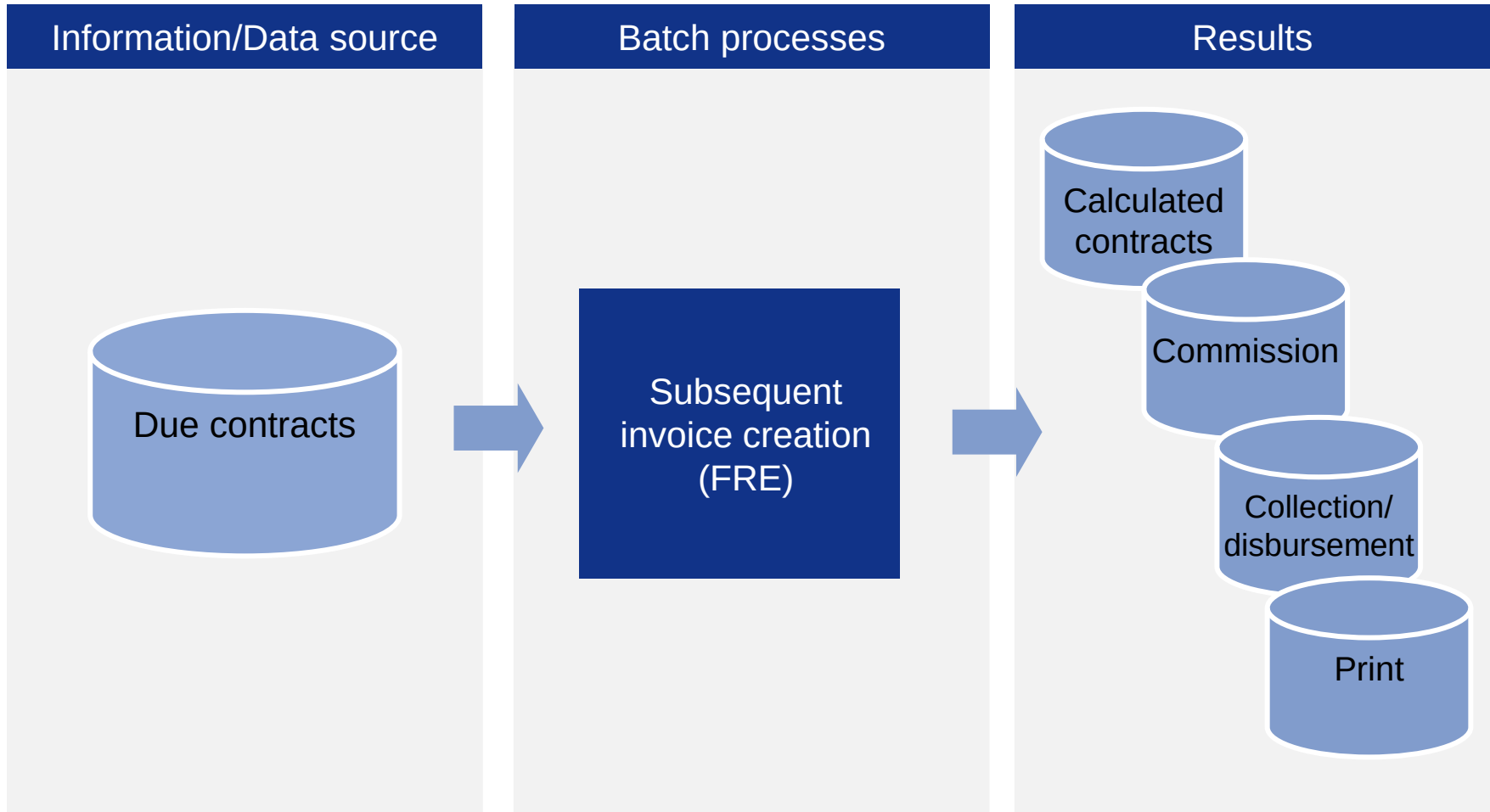
Collection/disbursement batch processes are running before, during and after online



As a cross-cutting function, collection/disbursement services are available for all batch processes during the entire operating day.

Monthly special action parallel to online - FRE

Invoices are created for the contracts due during the settlement period.



Acceptance of due contracts and if necessary value adjustments as well as invoicing

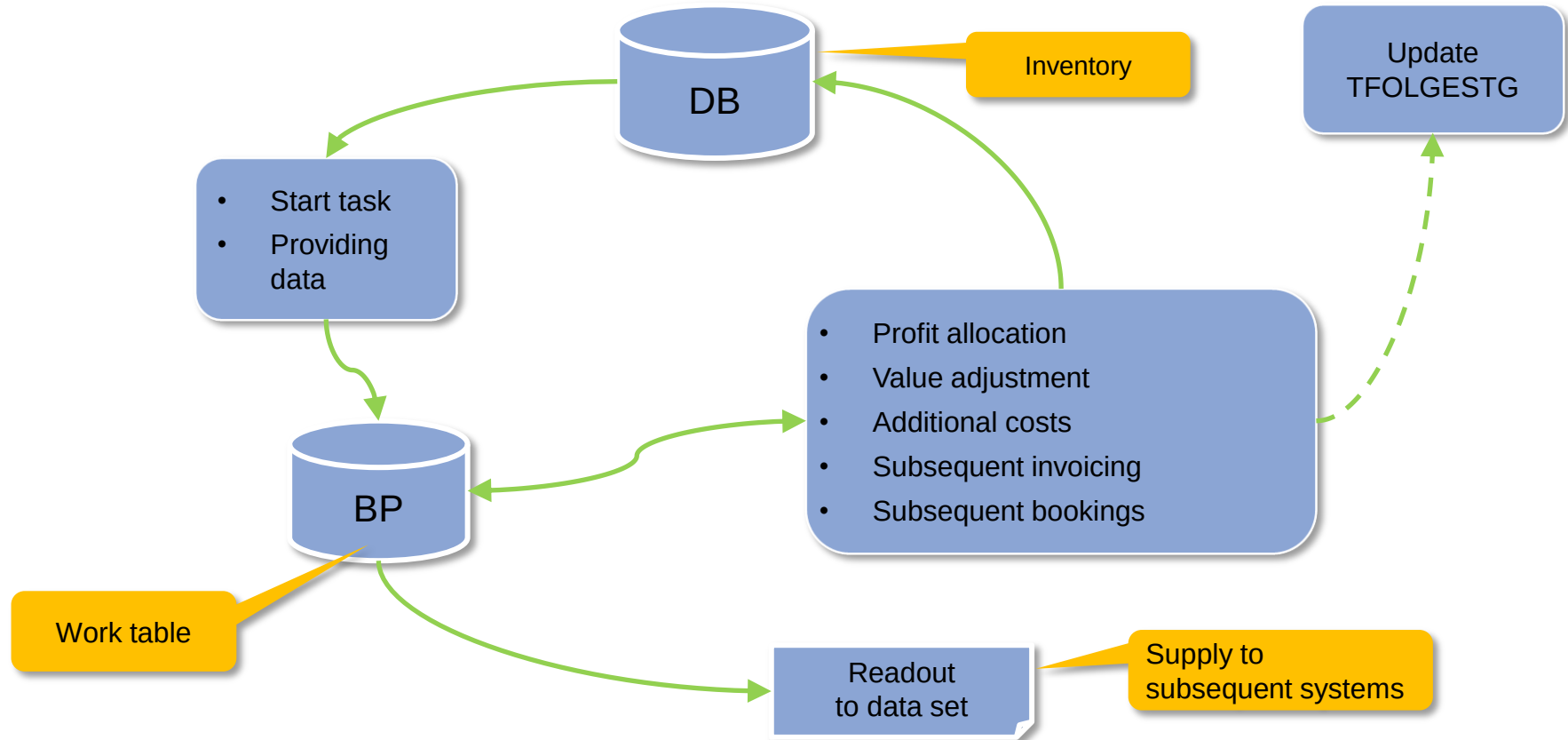
Monthly subsequent invoice creation (FRE)

Selection of due contracts based their payment frequency

Selection	Determining relevant contracts and provision for processing; Subsequent invoicing runs for the entire contract inventory.
Processing a contract due in the subsequent month	Profit allocation for life insurance
	Premium, value adjustment and reclassification based on index adjustments
	Specify subsequent amount, create bookings
	Collect print data, creating printed documents
Completion	Database update
	Set a new subsequent month

Work steps of subsequent invoice creation (FRE)

Representation of the sequence of individual work steps during FRE



Basic work steps are specified by ABS Core and cannot be changed. However, customer-specific extensions are possible.

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ABS batch service concept

ABS batch service concept offers number of preferences and supports the 24x7 operation.

The following targets are the main focus

Load and time-dependent scaling of batch processing during the online operation

Standardisation in program controls → easier to maintain

Reduction of program aborts due to standardised error handling

Parallel activities of processing and processes → Reduction of throughput times

Batch processing parallel to online operation allows for → short waiting periods

Online monitoring → controlling the processes depending on the work load

Generated batch services are providing the following advantages

Standardised features make it easier to develop, test, maintain and locate errors

Genuine “ON ERROR” units → if I have a problem, it is addressed in the batch main program

Final return code handling → Status message for the processing course

Rollback for test run → Withdrawal of database changes

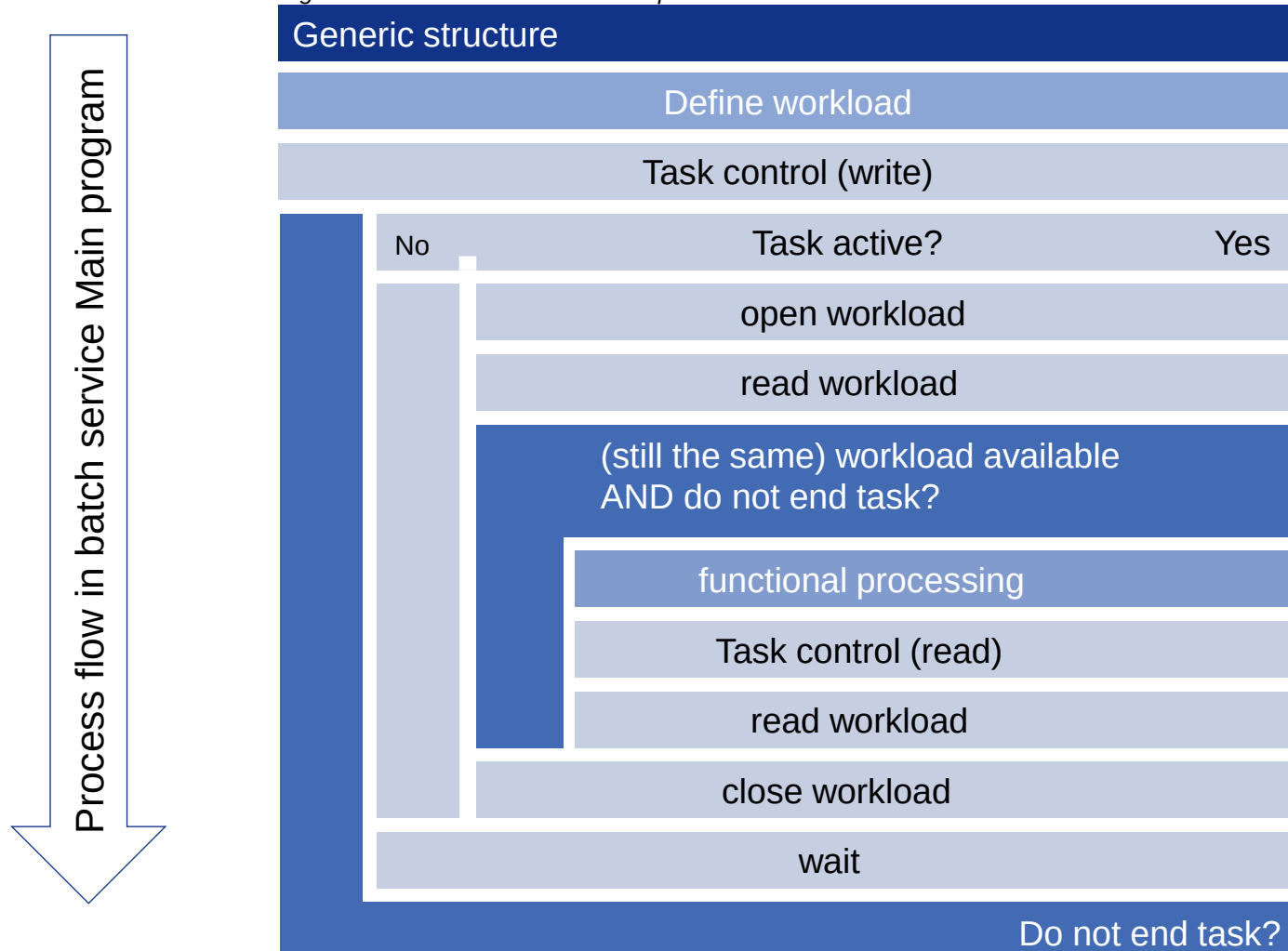
Support parallelism

Provision of grand totals → may only take place in the main program

Support of selective business object processing (controllable from the outside)

Standardization of program control

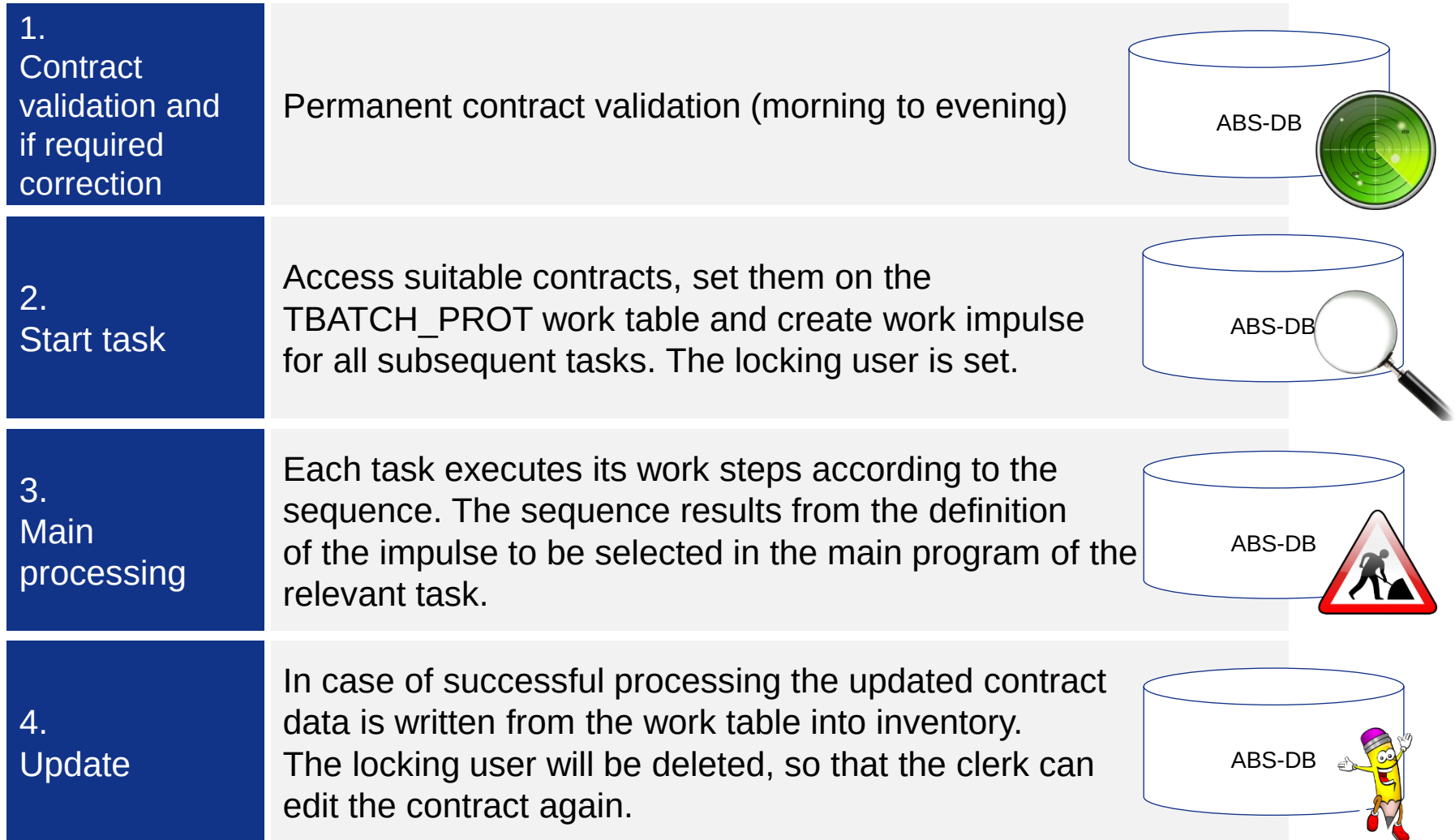
The Nassi- Schneidermann diagram illustrates the structural sequence of a batch service



A batch service main program is generated and is only adapted manually for the workload definition.

Processing steps of an ABS batch application - overview

Schematicel representation of the principle process of an entire ABS batch application using the example of the policy batch.



1. Contract validation and correction, if required

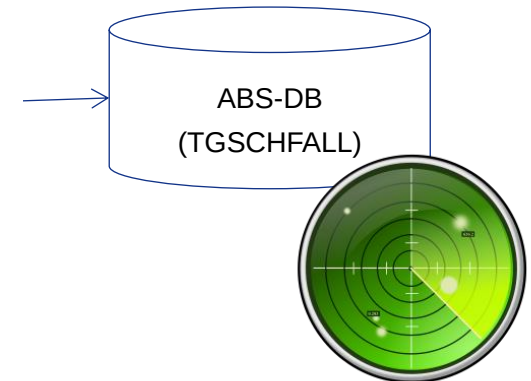
Technical contract validation and correction, if required, with subsequent provision for the start task

Contract checkers run continuously and scan new or changed contracts for data consistency

If “OK”, mark as “to be policy processed” (via GSTUEBRNZEIT)

If “not OK”, try to correct or create a process slip

```
AS-0184417155 STADIUM `3` GPSTATUS ,E` GSTUEBRNZEIT ,0001-...`
AS-0184417162 STADIUM ,2` GPSTATUS ,A` GSTUEBRNZEIT ,0001-...`
AS-0184428601 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0005-...`
AS-8117616174 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0005-...`
AS-8117636147 STADIUM ,2` GPSTATUS ,W` GSTUEBRNZEIT ,0001-...`
AS-8117637658 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0005-...`
```



2. Start task policy processing

Copying the operative tables in the work tables and creating impulse for the subsequent jobs

Search for applications to be policy processed based on GSTUEBERNZEIT + GPSTATUS

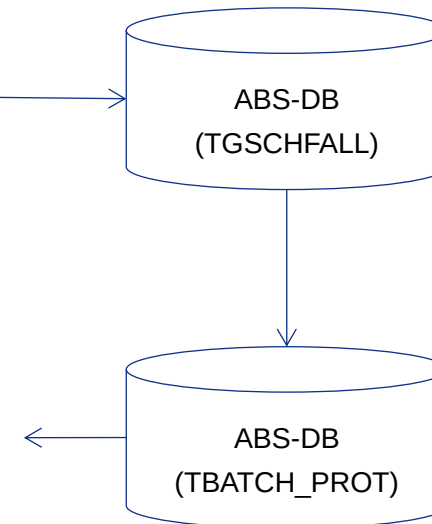
Set current work status in the work table TBATCH_PROT

Move all subsequent jobs with processing flag "O" for open to TBATCH_PROT

Set TGSCHFALL.SPERRUSER = „#VPZJOST“

```
AS-0184417155 STADIUM `3` GPSTATUS ,E` GSTUEBRNZEIT ,0001-...`
AS-0184417162 STADIUM ,2` GPSTATUS ,A` GSTUEBRNZEIT ,0001-...`
AS-0184428601 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0002-...`
AS-8117616174 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0002-...`
AS-8117636147 STADIUM ,2` GPSTATUS ,` GSTUEBRNZEIT ,0001-...`
AS-8117637658 STADIUM ,2` GPSTATUS ,F` GSTUEBRNZEIT ,0002-...`
```

GF#	GFAKTIVITAET	STRUKTURNAME	VERARBEITUNGSKZ
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOST	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOAN	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOVR	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOWP	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDSJO01	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDDJOPD	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJORU	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	UBHJOVS	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOV	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJODB	O



3. Main processing

Task-specific processing steps with appropriate acknowledgement of own work results for the subsequent task

Each task searches for its name in TBATCH_PROT (e.g. UBHJOVS). If processing flag = “O” and the parent = “E” then begin own processing

After successful processing of the work set the field VERARBEITUNGSKZ is set to „E“. If an error occurs it's set to „X“.

If an error occurs the actual contract remains unaffected and consistent in the inventory. Everyday there is a new attempt to process the contract.

GF#	GFAKTIVITAET	STRUKTURNAME	VERARBEITUNGSKZ
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOST	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOAN	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOVR	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOWP	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDSJO01	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDDJOPD	X
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJORU	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	UBHJOVS	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOFV	O
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJODB	O



4. Update of inventory

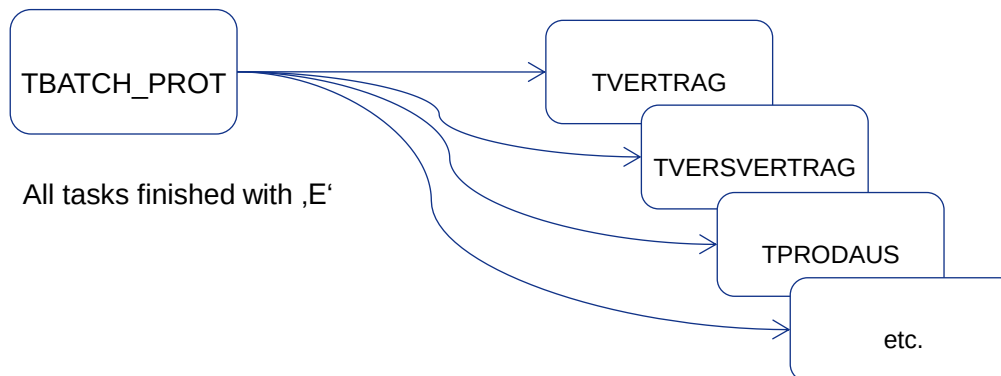
After successful total processing, the entire business object is updated and unlocked

If every task has acknowledged its work with “E”, DB update can start and transfer the current status of the contract from the work table to the inventory tables.

From this time, the new contract is visible in the client and can be re-changed.

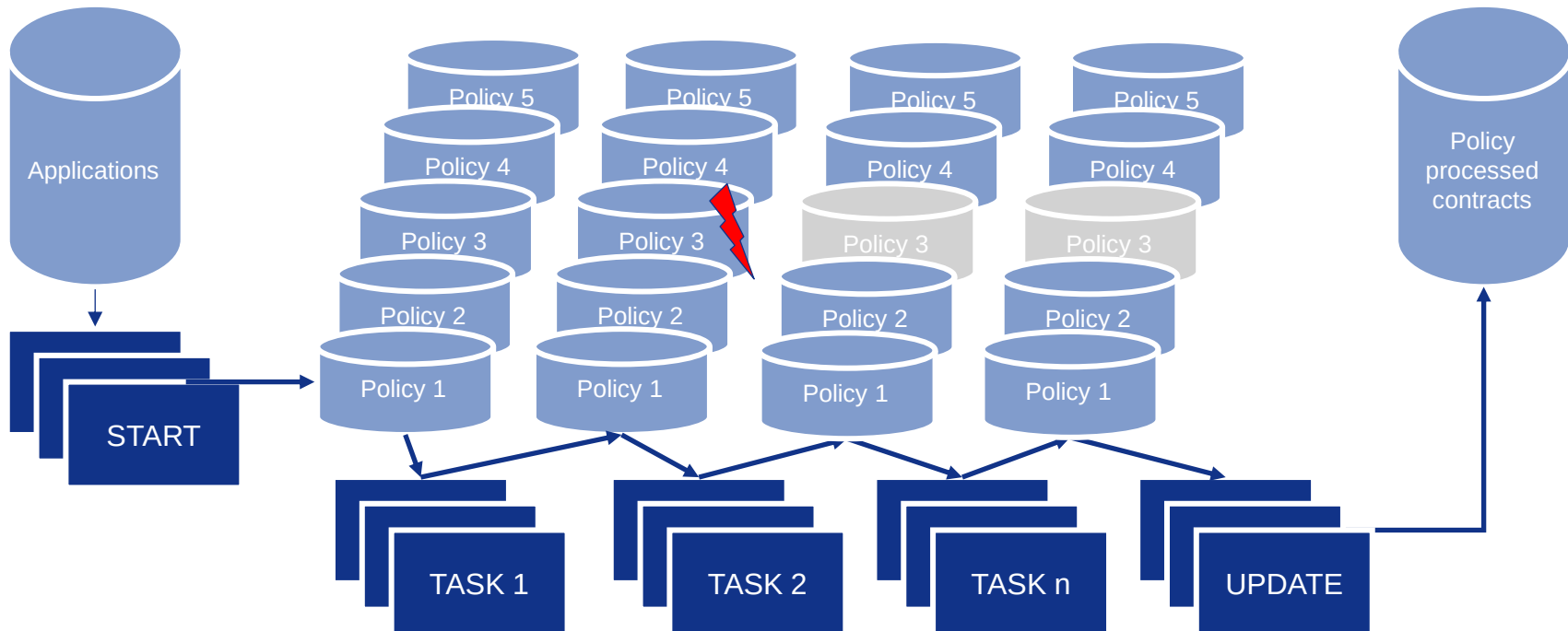
The business object is released by removing the locking user „#VPZJOST“.

GF#	GFAKTIVITAET	STRUKTURNAME	VERARBEITUNGSKZ
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOST	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOAN	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOVR	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOWP	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDSJO01	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	PDDJOPD	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJORU	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	UBHJOVS	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJOFV	E
57d80292-8af9-11e2-beb9-070bbf36034f	POLDF	VPZJODB	E



The batch services operate on a work table

Parallel assembly line processing of contracts without mutual interference



Much like on an assembly line, continuous production takes place at each station.

A completed assignment is acknowledged and processed by a subsequent program.

Depending on the process load, several jobs accept work list at a station.

At the end of the chain, acceptance takes place from the work table in the operational DB.

Grand total protocol

Protocol for Program **VPZJOST**

Compile: 12.JUN.14 18.20.10

Start: 2015-03-12-07.00.59.991**Stop: 2015-03-12-18.00.48.503**

Duration: 10.59.48.512

CPU-time: 00.00.14.860

STEP-CPU-time: 00.14.33.858

JOBNAME: RNV0000W

JOBID: JOB01524

STEPNR: 4

SYSID: APO0

Returncode=0005

Control statement:

VPZJOST ,RNV0000W ,01,XRACE,00100, ,00000, ,003000

S,2015-03-12,2015-03-01

A,2015-03-12,2015-02-01

L,2015-03-12,2015-03-01

Totals:

01	29.717,00	business cases read
02	725,00	processed amendment notices
03	0,00	not processed amendment notices
04	26.981,00	processed applications
05	5,00	not processed applications
06	0,00	not found contracts
07	624,00	applications not yet for processing
08	3,00	second change
09	1.379,00	applications with other escalation reasons
10	0,00	number of deadlocks
11	5,00	technically not processed business objects
12	2.006,00	functional not processed business objects
13	27.706,00	proccesed business cases for VPZJOAN
14	0,00	-----
15	0,17	Sec. average duration per GF



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Summary

What should we keep in mind?

According to this module, you should be able to make a statement by taking into consideration the following points

What distinguishes the ABS batch mode?

How does exchange take place between the ABS Client and the ABS database?

Why is the business object manager required?

What distinguishes the ABS operating day and what are its basic features?

How are the basic processes setup in the batch?

What are the special features of ABS Batch architecture?

Example test question

Quiz navigation

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23	24	25	26	27	28	29	30	31	32	33
34	35	36	37	38	39	40	41	42	43	44
45	46	47	48	49	50	51	52	53	54	55
56	57	58	59	60	61	62	63	64	65	66
67	68	69	70	71	72	73	74	75	76	77
78	79	80	81	82	83	84	85	86	87	88
89	90	91	92	93	94	95	96	97	98	99
100	101	102	103	104	105	106	107	108	109	110
111	112	113	114	115	116	117	118	119	120	

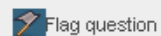
Finish attempt ...

Time left 1:23:55

Question 110

Not yet answered

Marked out of 1.00


[Next](#)

11-0002: Which of the following terms are used in the ABS Batch environment?

Select one or more:

- ☐ A. Batch Service
- ☐ B. Batch programs
- ☐ C. Batch Framework
- ☐ D. Batch Job
- ☐ E. Batch WebService

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